

NetGuard Pro — README

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Centrally managed network-traffic inspection and threat-detection engine.

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1. Overview / Executive Summary

NetGuard Pro is an enterprise-grade network security and monitoring platform developed by NetGuard Solutions. It inspects East-West and North-South traffic, detects lateral-movement attempts, and provides centralised security management for on-premises, cloud, and hybrid environments.

Key capabilities include real-time threat detection, automated firewall policy enforcement, VPN management, SIEM integration, role-based access control, and customisable compliance reporting.

Audience: IT managers, architects, security officers, and executive stakeholders.

2. Security Posture & Compliance

NetGuard Pro is designed to align with major security frameworks and regulatory standards:

- Authentication & Authorisation: Role-Based Access Control (RBAC), MFA, SSO via SAML 2.0 / OIDC.
- Data Protection: AES-256 encryption at rest; TLS 1.3 in transit. Logs are anonymised by default.
- Compliance Frameworks: ISO 27001, NIST CSF, SOC 2 Type II, GDPR, PCI-DSS.
- Supply Chain Security: Software Bill of Materials (SBOM) available upon request; all dependencies scanned via SCA.

Audience: Security officers, compliance teams, and auditors.

3. Architecture & Components

NetGuard Pro follows a split control-plane / data-plane architecture:

- Management Server (Control Plane): Central console for policy management, dashboards, reporting, and user administration.
- Sensors / Agents (Data Plane): Lightweight probes deployed on network segments, endpoints, or as virtual appliances.
- Communication Channels: REST API (HTTPS), encrypted RPC for sensor-to-server communication, Kafka-based event bus for SIEM integration.

A full architecture diagram is available at docs.netguardsolutions.com/architecture.

Audience: Architects, DevOps engineers, and support teams.

4. System Requirements

Minimum

- OS: Windows Server 2019 / Ubuntu 20.04 LTS or later
- CPU: 4-core, 2.0 GHz | RAM: 8 GB | Storage: 50 GB | Network: 1 Gbps NIC

Recommended

- CPU: 8-core, 3.0 GHz | RAM: 16 GB | Storage: 200 GB SSD | Network: 10 Gbps NIC

Audience: DevOps engineers and sysadmins.

5. Deployment & Installation

Prerequisites: Ensure the following ports are open: 443 (HTTPS), 8443 (sensor communication), 514 (syslog).

1. Download the installer from the NetGuard Solutions portal.
2. Verify the SHA-256 checksum against the published hash.
3. Run as Administrator (Windows) or with sudo (Linux).
4. Follow the setup wizard to configure network interfaces and initial policies.
5. Register sensors: navigate to Sensors > Add New and follow the onboarding wizard.
6. Enter your license key when prompted and restart the system.

For silent/unattended installations and Kubernetes deployments, see docs.netguardsolutions.com/deployment.

Audience: DevOps engineers, sysadmins, and deployment engineers.

6. Configuration & Tuning

Primary configuration file: `/etc/netguardpro/config.yaml` (Linux) or `C:\NetGuardPro\config.yaml` (Windows).

Key tunable parameters:

- `detection_sensitivity`: LOW | MEDIUM | HIGH — adjust to reduce false positives in trusted environments.
- `policy_rules`: Define whitelists, blacklists, and custom detection signatures.
- `network_segments`: Specify monitored interfaces and traffic selectors (DMZ, LAN, cloud-VPC).
- `log_retention_days`: Default 90 days; adjust per compliance requirements.

Example configuration profiles for DMZ, internal LAN, and cloud-VPC are available in [/examples/configs/](#).

Audience: Security engineers, network admins, and support teams.

7. Usage & Workflows

Common Operations

- Launch console: `netguardpro --start` | Web UI: `https://<server-ip>:443`
- Create a firewall policy: Security > Firewall Rules > New Rule
- Run a traffic baseline scan: Monitoring > Baseline > Run Now
- Review alerts: Dashboard > Active Alerts

API Quick Reference

GET `/api/v1/alerts` — retrieve active alerts

POST `/api/v1/policies` — create a new policy rule

GET `/api/v1/sensors/status` — check sensor health

Full API documentation with example payloads at docs.netguardsolutions.com/api.

Audience: Day-to-day security and network operators.

8. Monitoring, Logging & Troubleshooting

Log Locations

- System logs: `/var/log/netguardpro/system.log`
- Audit logs: `/var/log/netguardpro/audit.log`
- Detection events: `/var/log/netguardpro/detections.log`

SIEM Integration

- Supports syslog (RFC 5424), Splunk HEC, and ELK Stack via Logstash.
- Pre-built dashboards for Splunk and Kibana available at docs.netguardsolutions.com/siem.

Health Checks

- Status API: `GET /api/v1/health`
- Heartbeat interval: configurable (default 60 seconds).

Common Errors

- `SENSOR_UNREACHABLE`: Verify port 8443 is open and sensor service is running.
- `LICENSE_EXPIRED`: Contact licensing@netguardsolutions.com.
- `HIGH_MEMORY_USAGE`: Reduce `detection_sensitivity` or increase RAM allocation.

Audience: Support teams, NOC, and SRE engineers.

9. Security Hardening & Best Practices

Management Server

- Restrict SSH access to management IPs only; disable root login.
- Enable OS-level firewall; allow only ports 443, 8443, and 514.
- Rotate TLS certificates every 12 months; use certificates from a trusted CA.

Sensors / Agents

- Deploy sensors with least-privilege service accounts.
- Isolate sensor network interfaces from user traffic where possible.

Credentials & Keys

- Rotate API keys every 90 days.
- Store secrets in a vault solution (HashiCorp Vault, AWS Secrets Manager).
- Enable MFA for all administrator accounts.

Audience: Security engineers and auditors.

10. Supported Environments & Integrations

Platforms

- OS: Windows Server 2019/2022, Ubuntu 20.04/22.04, RHEL 8/9, Debian 11/12
- Cloud: AWS, Microsoft Azure, Google Cloud Platform
- Containers: Kubernetes (Helm chart available), Docker
- Hypervisors: VMware ESXi, Microsoft Hyper-V, KVM

Integration Points

- Identity Providers: SAML 2.0 (Okta, Azure AD), OIDC
- SIEM / SOAR: Splunk, IBM QRadar, Microsoft Sentinel, Palo Alto XSOAR
- Ticketing: Jira, ServiceNow
- Asset Inventory: ServiceNow CMDB, Qualys

Audience: IT managers and integration teams.

11. Support & Escalation Procedures

When contacting support, please include: software version, OS version, relevant log excerpts, and steps to reproduce.

- L1 Support (general): support@netguardsolutions.com | Response: 8 business hours
- L2/L3 Support (technical): techsupport@netguardsolutions.com | Response: 4 business hours
- Security-Critical Incidents: security@netguardsolutions.com | Response: 1 hour
- Support Portal: <https://support.netguardsolutions.com>
- Documentation: <https://docs.netguardsolutions.com>

Audience: Support teams and incident responders.

12. Roadmap & Known Limitations

Upcoming Features (v1.2 — Q3 2026)

- SAML SSO configuration wizard.
- IPv6 full deep packet inspection support.
- XLSX export for scheduled reports.
- Dashboard performance improvements for 500+ monitored devices.

Known Limitations (v1.1.0)

- IPv6 deep packet inspection not fully supported in all environments.
- Dashboard latency with more than 500 simultaneous monitored devices.
- SAML SSO requires manual configuration (wizard in v1.2).

Audience: IT managers and architects planning capacity.

13. Changelog

v1.1.0 — March 2026

- Added Role-Based Access Control (RBAC) module.
- Improved dashboard performance for large topologies.
- Fixed VPN tunnel reconnection issue after system sleep.

v1.0.0 — January 2026

- Initial release.

14. License & Legal Notices

- License: Commercial — NetGuard Solutions Proprietary License v2.0.
- Copyright: © 2026 NetGuard Solutions. All rights reserved.
- Export Control: This software is subject to export control regulations. Distribution outside the EU/EEA requires prior written consent.
- Third-party components and their licenses are listed in THIRD_PARTY_NOTICES.txt.

Audience: Legal, procurement, and auditors.

15. Contributing

NetGuard Pro welcomes contributions for documentation improvements, bug reports, and feature suggestions.

- Bug reports: github.com/netguard/netguardpro/issues
- Documentation improvements: fork the repo and open a pull request.
- Feature requests: feedback.netguardsolutions.com
- Security vulnerabilities: report privately to security@netguardsolutions.com — do NOT open a public issue.
- Coding standards and branch strategy: see CONTRIBUTING.md in the repository.

Audience: Partner developers and internal R&D teams.